UNIVERSITATEA TEHNICĂ
DIN CLUJ-NAPOCA

# COMPUTER AIDED DESIGN PROJECT

# VOLTAGE STABILIZER WITH OVERCURRENT PROTECTION

# 2025

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# 1. Project requirements

Design a voltage stabilizer where the output voltage is adjustable in the range Vout_min 7V and Vout_max 20V, the output current is limited to the maximum value Io_lim 1.5A, knowing that a constant voltage V_in 25V is applied to the circuit input.

# 2. Technical details

1. The output voltage level must be maintained between 7V and 20V.
2. The output current must not exceed 1.5A.
3. The load resistance that varies to observe the correct current limitation is between 0Ω and 100kΩ.
4. The circuit is powered from a 25V constant voltage source.
5. Regulation transistor: Q2N2222.
6. Protection transistor: Q2N3055.
7. The resistors used for the design are part of the E96 series with ±1% tolerances.
8. Operational amplifier: TL082.
9. The voltage reference is designed using the 5.1V Zener diode 1N4733/1N4733A.

# 3. Block diagram

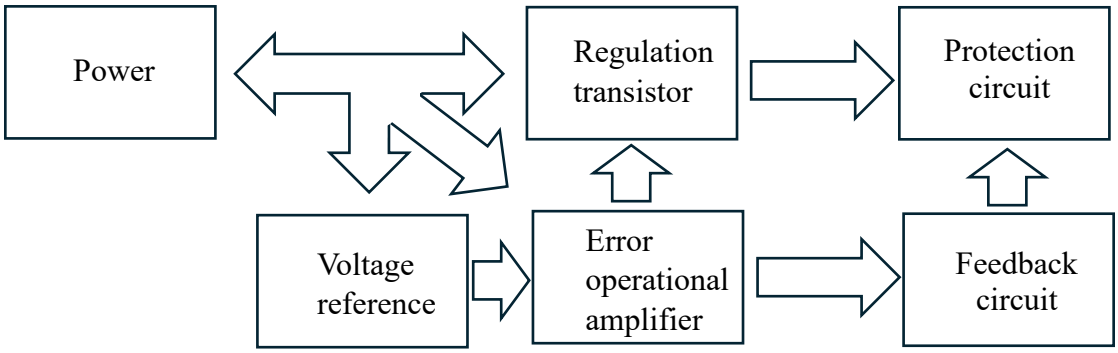


Figure 1. Block diagram

# 4. Electrical schematic

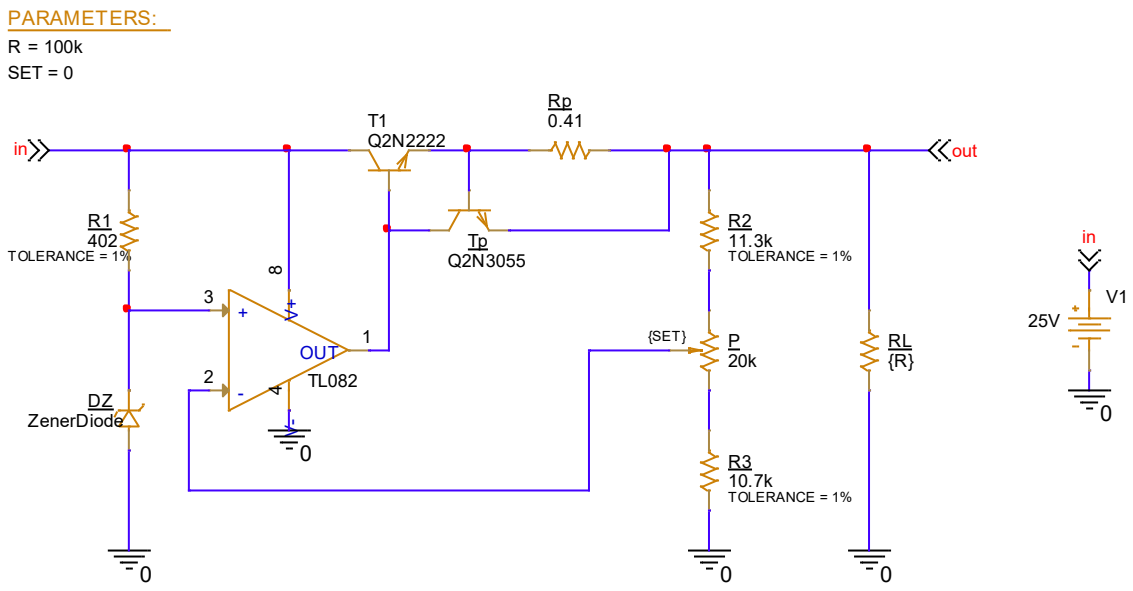


Figure 2. Electrical schematic

# 5. Detailed calculation and design of the circuit

## 5.1. Voltage reference

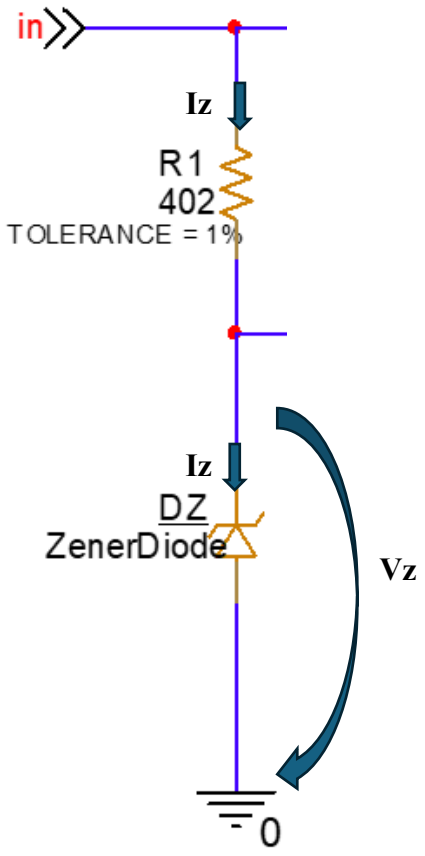


Figure 3. Voltage reference

The voltage reference in Figure 3 is an essential part of the circuit because it allows us to stabilize the output voltage. For the voltage reference we need a Zener diode and a resistor to protect the diode.

### 5.1.1. R1 value calculation

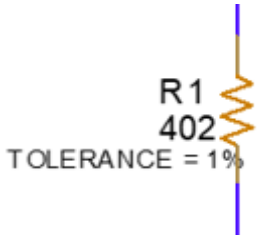


Figure 4. Resistor R1

From the datasheet of the 1N4733 Zener diode we deduce that the diode stabilizes at a voltage of 5.1V (Vz) for a current of 49mA (Iz). Thus, we will calculate the resistance R1 using the relationship between voltage, current and resistance:

$$R_1 = \frac{V_{in} - Vz}{Iz} = \frac{25V - 5.1V}{49mA} = 406.12\Omega \quad (1)$$

In the E96 series of standard resistors we do not find this value, but the closest real value is 402Ω, but then the current through the diode and the stabilization voltage will change according to relationship (1) as follows:

$$Iz = \frac{V_{in} - Vz}{R_1} = \frac{25V - 5.1V}{402\Omega} = 49{,}5mA$$

$$V_{in} - V_z = Iz \cdot R_1 => V_z = V_{in} - R_1 \cdot Iz = 25V - 402\Omega \cdot 49.5mA = 5.101V$$

Although we have the current of 49.5mA in this case, this small difference changes the stabilization voltage value very little, from 5.1V to 5.101V, being a negligible difference. Consequently, we can replace the value of resistor R1 with 402Ω for a real design.

However, if we want to obtain that resistance of exactly 406.12Ω, we can use the following configuration of resistors from the E96 series:

402Ω în serie cu 4.12Ω ⇒ $406.12\Omega$

### 5.1.2. Zener diode modeling

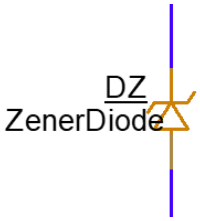


Figure 5. Zener diode

With the help of the datasheet of the Zener diode we want to use, we are able to model a diode suitable for our application.

To model the diode in Figure 5, we need all the necessary data from the datasheet to determine the current-voltage characteristic at forward and reverse bias.

The current-voltage characteristic at forward bias involves the variation of the current through the diode as a function of the voltage across the diode. In forward bias, we know that a Zener diode behaves identically to a regular diode, having a threshold voltage of approximately 0.7V, as can be seen in figure 6 taken from the diode's datasheet.

In OrCAD, the diode is modeled at a temperature of 27degC, and in the datasheet we obtain the values at a temperature of 25degC. According to Figure 6, the differences between the characteristics are quite small and compared to large variations between temperatures, therefore, the diode behavior at 27degC will be similar to that at 25degC. The diode has been modeled in Pspice Model Editor.

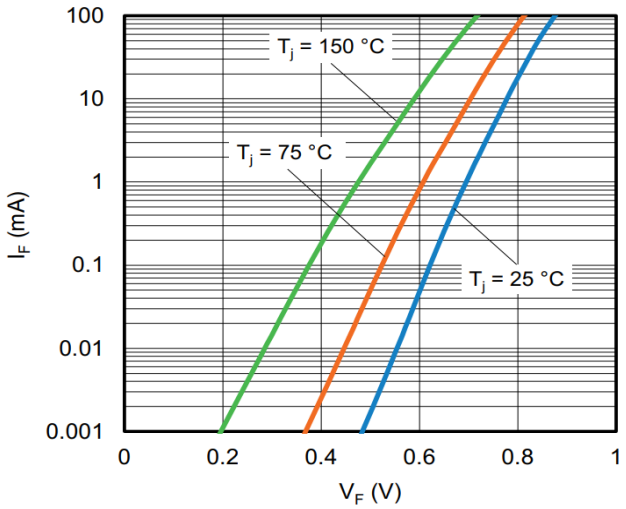


Figure 6. Forward bias characteristic at different temperatures

The diode datasheet also states that the diode must have a maximum forward bias of 200mA at a voltage of 1.2V. Now that we know all this information, we can fill in the Forward Current table in OrCAD required for the forward bias of the diode.

Next, for the reverse bias current-voltage characteristic we need a little more data from the datasheet to help us inform OrCAD about the Reverse Leakage and Reverse Breakdown part of the characteristic.

Firstly, for the Reverse Leakage part, where we basically have those small values of current before the Zener diode starts to stabilize, we deduce from the datasheet that at a reverse current (IR) lower than 10uA we have a reverse voltage (VR) of 1V, therefore, two correct sets of values (IR, VR) for this region could be (0.1uA, 1V) and (10uA, 3V). We fill these values in the Reverse Leakage table in OrCAD.

Secondly, for the Reverse Breakdown part we need the Zener voltage of 5.1V and the Zener current of 49mA. In addition to these values, OrCAD also needs an impedance Zz, which for our diode is < 7 Ω, according to the datasheet. Using the PSpice model of a 1N4733 diode already modeled in OrCAD, we realize that at a current of 20mA we have an impedance Zz equal to 2.2 Ω. So, we calculate Zz for our diode using the simple rule of three, considering the stabilization slope on the reverse characteristic to be linear:

$$\left.\begin{array}{l} 2.2\Omega \ldots 20mA \\ Zz \ldots\ldots 49mA \end{array}\right\} \Rightarrow \quad Zz = \frac{2.2\Omega \cdot 49mA}{20mA} = 5.39\Omega$$

This obtained value corresponds to the requirements in the datasheet, therefore we consider it a valid value. With this information, we now have all the necessary data for the reverse bias current-voltage characteristic, and the diode is now fully modeled.

## 5.2. Error amplifier

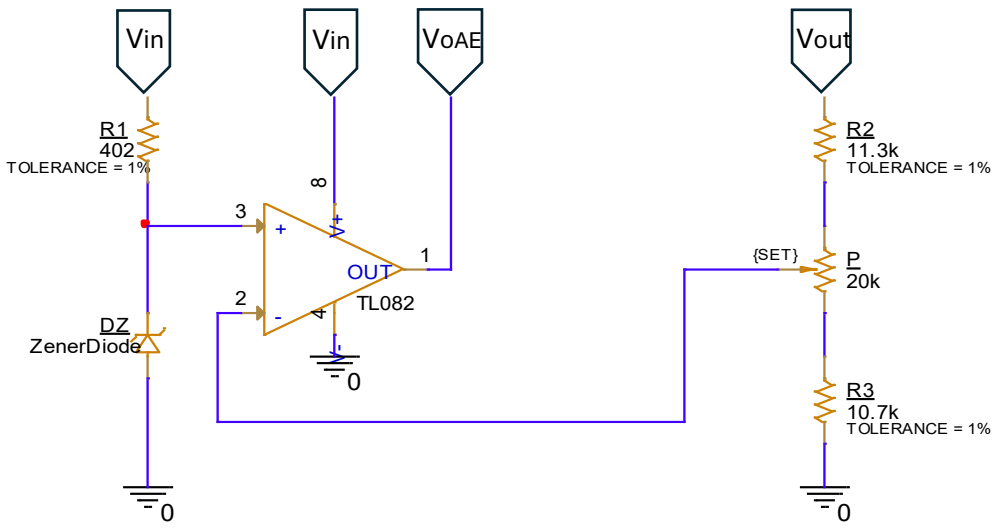


Figure 7. Error amplifier

*VoAE* – error amplifier output voltage

The TL082 error amplifier in Figure 7 is essential for our circuit because it allows us to connect the voltage reference connected to the non-inverting terminal to the feedback circuit connected to the inverting terminal. We used this operational amplifier because it can handle a supply voltage of 25V. Due to the negative feedback, it will result that v+ = v- (2).

$$v^+ = V_z\,, \qquad v^- = \frac{\alpha P + R_3}{R_3 + P + R_2} V_{out}$$

From (2) we get:

$$V_{out} = \frac{R_3 + P + R_2}{\alpha P + R_3} V_z \quad (3)$$

Thanks to the potentiometer in the feedback circuit, we will be able to divide the output voltage V_out in the established range, depending on the position of the cursor α (named SET on the circuit) of the potentiometer P (figure 7). From relation (3) it can be seen that the value of the output voltage depends both on the position of the potentiometer cursor and the reference voltage.

The output voltage variation range must be between 7 V and 20 V, which means that when the potentiometer slider is at maximum, we will need to obtain the minimum value of the output voltage, and when the potentiometer slider is at minimum we will need to obtain the maximum voltage, according to relation (3).

$\alpha = 1 \Rightarrow 7V = \frac{R_3+P+R_2}{P+R_3} \cdot 5.1V \Rightarrow \frac{R_3+P+R_2}{P+R_3} = 1.37$ (4)

$\alpha = 0 \Rightarrow 20V = \frac{R_3+P+R_2}{R_3} \cdot 5.1V \Rightarrow \frac{R_3+P+R_2}{R_3} = 3.92$ (5)

From (4) and (5) we get:

$$\begin{cases} \frac{R_3+P+R_2}{R_3+P} = 1.37 \\ \frac{R_3+P+R_2}{R_3} = 3.92 \end{cases}$$

From (4) $\Rightarrow R_3 + P + R_2 = 1.37(R_3 + P)$ (6)

From (5) $\Rightarrow R_3 + P + R_2 = 3.92R_3$ (7)

Equalizing (6) and (7) we obtain the expression for R3:

$1.37(R_3 + P) = 3.92R_3$

$1.37R_3 + 1.37P = 3.92R_3$

$2.55R_3 = 1.37P$

$R_3 = 0.537P$ (8)

The expression for R2 is obtained equalizing (7) and (8):

$\Rightarrow P + R_2 = 2.92 \cdot 0.537P \Rightarrow R_2 = 0.568P$

For sizing, let $P = 20k\Omega$ $\Rightarrow$ $\begin{cases} R_2 = 0.568 \cdot 20k\Omega = 11.36k\Omega \\ R_3 = 0.537 \cdot 20k\Omega = 10.74k\Omega \end{cases}$

The value of the potentiometer P of 20kΩ is standard, but the calculated values for the resistors R2 and R3 are not, therefore, the value of R3 will be replaced with 10.7kΩ, and the value of R2 will be replaced with 11.3kΩ, both from the E96 series.

## 5.3. Regulation transistor and protection circuit

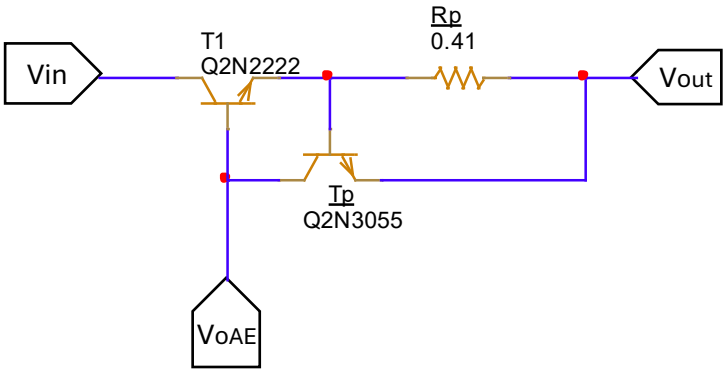


Figure 8. Regulation transistor and protection circuit

We find the value of Rp:

$$Rp = \frac{Vbe}{Io_lim} = \frac{0.65V}{1.5A} = 0.433\Omega \quad (9)$$

With this value of the resistance Rp obtained from relation (9) we have a limiting current of 1.5A only in the ideal case, when the voltage on the control junction of the transistor (Vbe) is exactly 0.65V. In OrCAD, however, this voltage drop is not exactly 0.65V, but smaller, approximately 0.61-0.62V, therefore the limiting current will only be around 1.44A in this case, according to the relation below:

$$Io_lim = \frac{0.62V}{0.433\Omega} \cong 1.44A$$

To increase the current value to 1.5A, we decrease the value of the resistor Rp to 0.41Ω. This value is not standard, but can be replaced with a standard one of 0.412Ω for an accurate design. For simulations we will keep its value at 0.41Ω.

$$Io_lim = \frac{0.62V}{0.41\Omega} \cong 1.5A$$

Transistor T1 in Figure 8 is a bipolar NPN control transistor that amplifies the current received from the output of the error amplifier and regulates the current through the base of the bipolar NPN power transistor Tp. Without transistor T1, the TL082 amplifier would have to directly control the current through the base of transistor Tp, which would not be ideal.

We chose the T1 Q2N2222 transistor because it is often used in electronic applications, which makes it easy to find and it supports enough current, 0.8A according to the datasheet, to be suitable for this application.

We chose the Tp Q2N3055 transistor because it is a power transistor and supports a high current (15A according to the datasheet) and fits perfectly in the configuration of a protection circuit.

The resistor Rp along with the transistor Tp form the protection circuit found in figure 8. This protection resistor works as a feedback for current limitation: it measures the current passing through it, and when the current exceeds Io_lim, the voltage on Rp increases and the protection transistor opens. As a result, the output voltage decreases due to the current limitation, thus protecting the circuit.

## 5.4. Load resistance

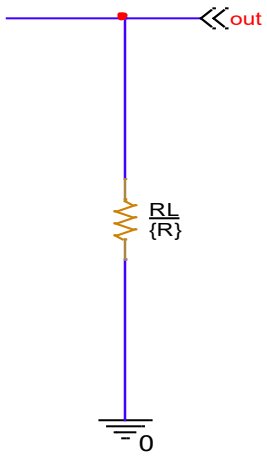


Figure 9. Load resistance RL

The load resistance RL in figure 9 helps us to observe, at different values , the correct current limitation and the voltage drop at current limitation. That is why the resistance does not have a fixed nominal value, but a value {R}, which signifies a variation of values in a certain range established during the simulation, in our case this range will be 0-100kΩ.

# 6. Simulations

## 6.1. Output voltage stabilization

To visualize the stabilization of the output voltage in the set range, we performed two different simulations with a Transient Analysis, one when SET = 0 (figure 11), and the other when SET = 1 (figure 12), and in both cases we showed the output voltage over a 100ms range, as can be seen in figure 10.

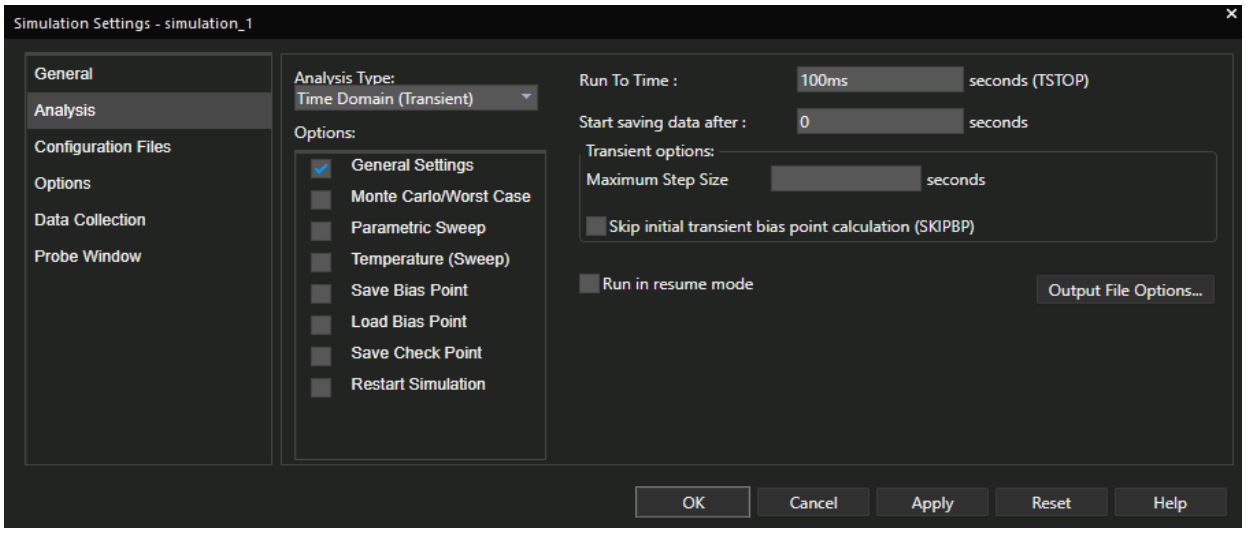


Figure 10. Transient simulation setup

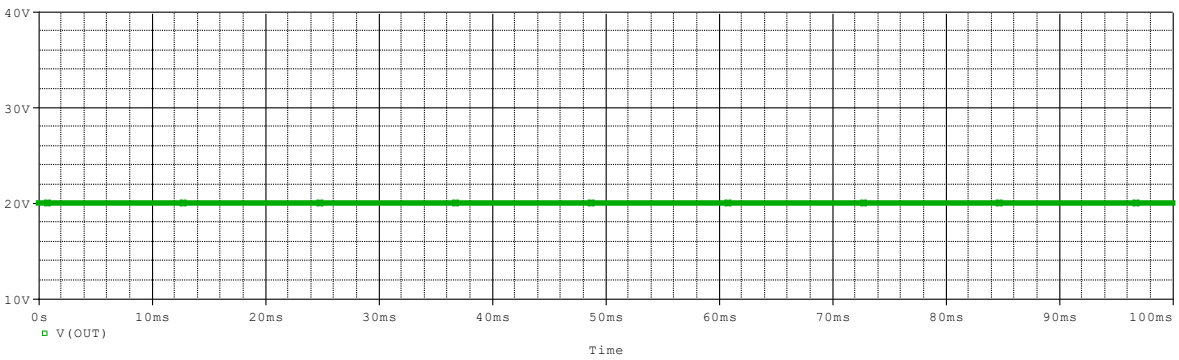


Figure 11. Maximum output voltage when SET = 0

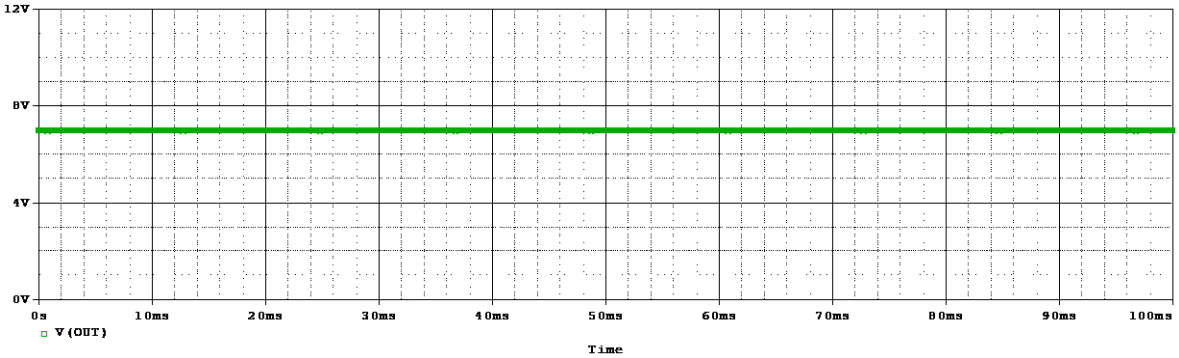


Figure 12. Minimum output voltage when SET = 1

## 6.2. Output voltage variation with cursor position

To see the variation of the output voltage depending on the position of the potentiometer cursor, we performed a Transient Analysis of 100ms, together with the Parametric Sweep option, where we vary the SET parameter of the potentiometer from 0 to 1 in steps of 0.1, as can be seen in figure 13.

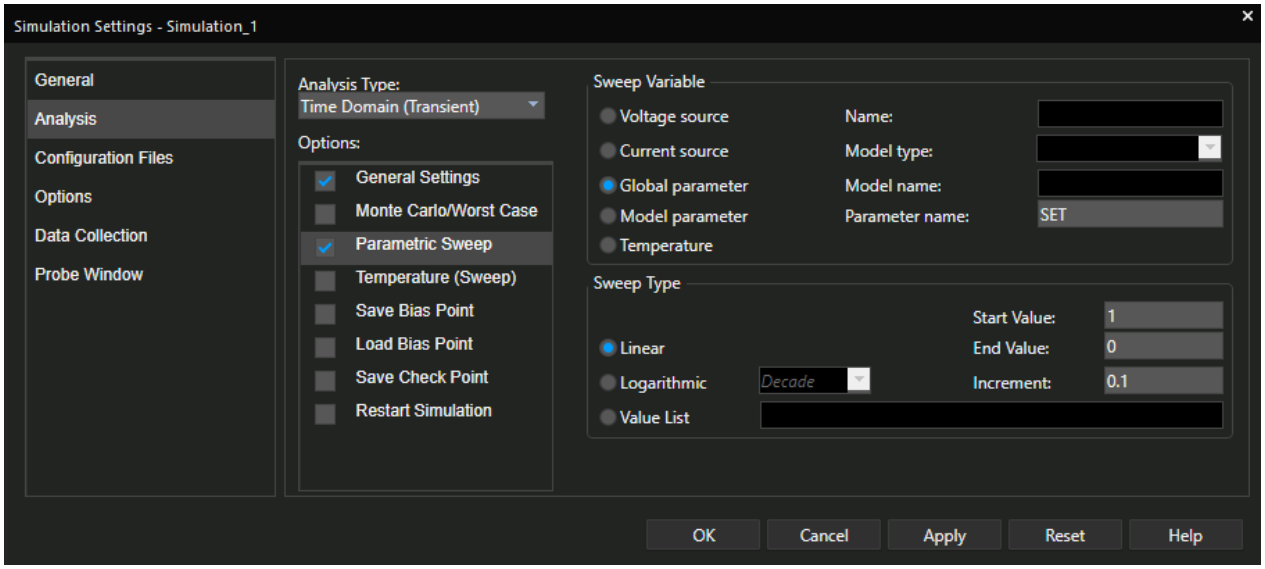


Figure 13. Simulation setup for parametric sweep

In figure 14 is the parametric analysis over time, which displays 11 different values of the output voltage at 11 different values, from 0.1 to 0.1, from 0 to 1, of the SET cursor.

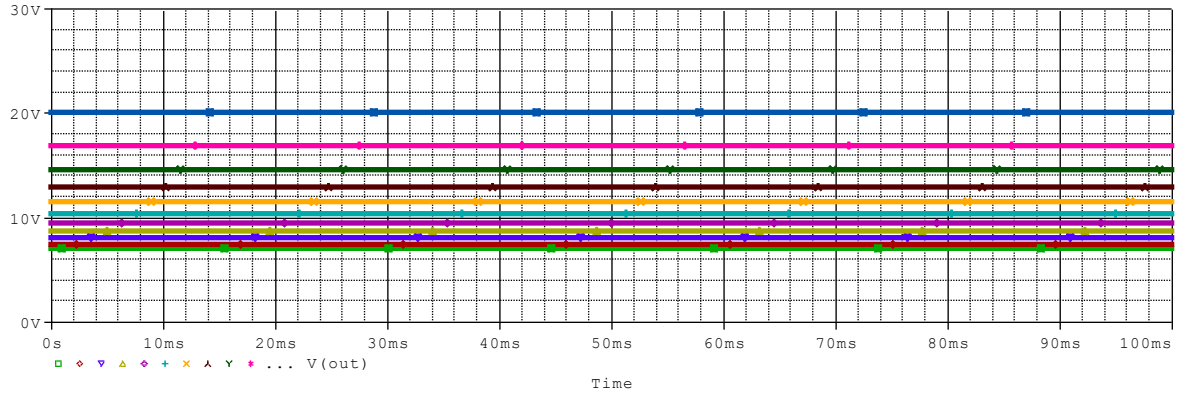


Figure 14. Output voltage for different values for SET

To be able to see the simulation results depending on the SET parameter, we performed a performance analysis and displayed the output voltage, as seen in figure 15.

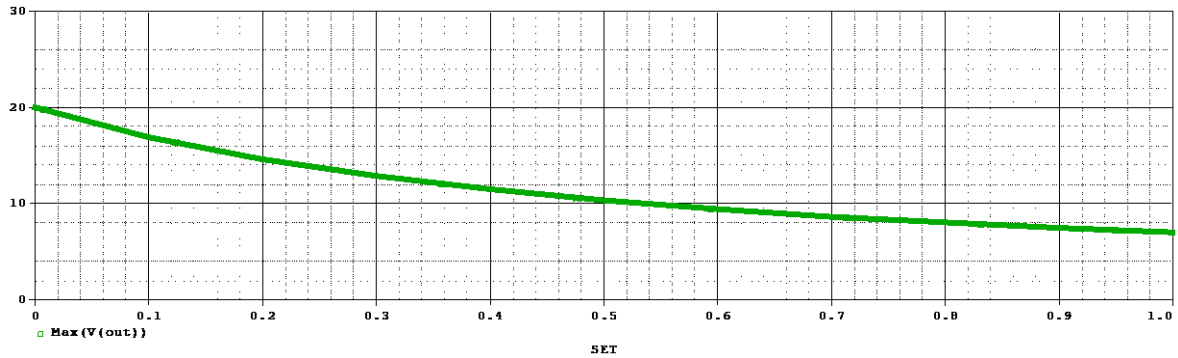


Figure 15. Output voltage across SET (SET between 0 and 1)

## 6.3. Voltage-current limiting characteristic

To determine the voltage-current limiting characteristic, we performed a Transient Analysis for 100ms, together with the Parametric Sweep option, where we varied the parameter R of the load resistance RL in the range 0-100kΩ, with a step of 100, as can be seen in figure 16.

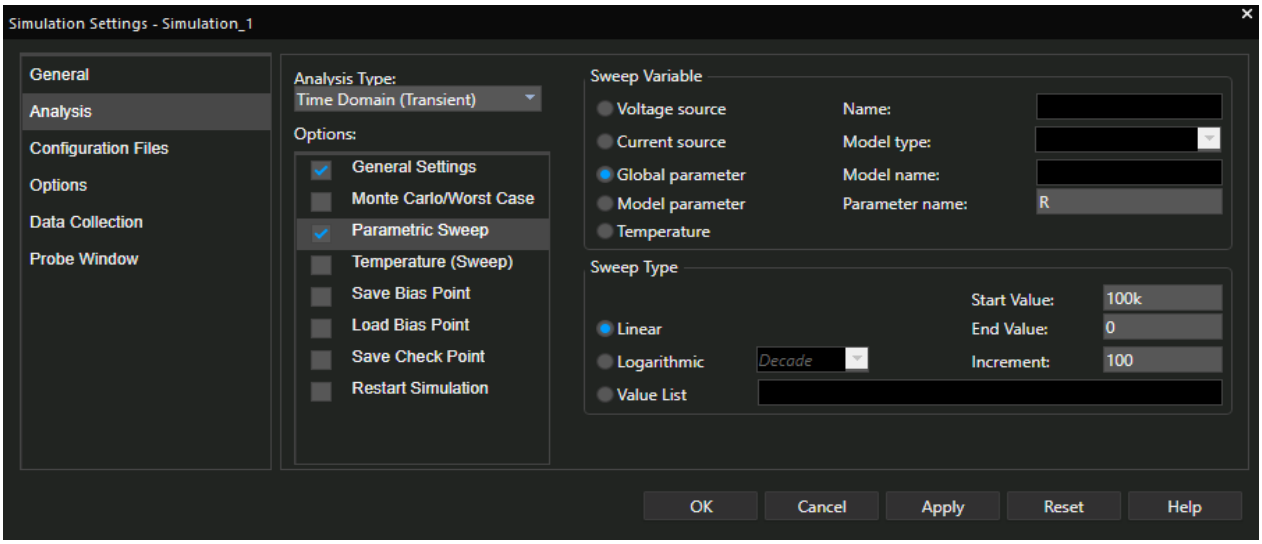


Figure 16. Simulation setup for parametric sweep for R

To visualize the simulation results, we also ran a performance analysis and displayed the output current and voltage across the RL load (here the output voltage is at 7V).

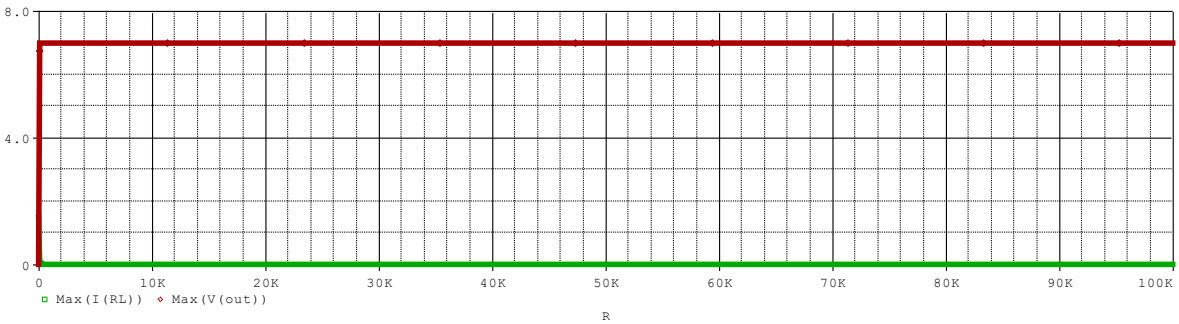


Figure 17. Voltage-current limiting characteristic

We can observe the correct operation from the characteristic. If the circuit output is shorted to grounf (zero output voltage), then the current through the load is limited to the limit current of 1.5A.

| Trace Color | Trace Name | Y1 | Y2 | Y1 - Y2 | | Y1(Cursor1) - Y2(Cursor2) | | 0.000 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | X Values | 0.000 | 0.000 | 0.000 | | Y1 - Y1(Cursor1) | Y2 - Y2(Cursor2) | Max Y | Min Y | Avg Y |
| CURSOR 1,2 | Max(I(RL)) | 1.5029 | 1.5029 | 0.000 | | 0.000 | 0.000 | 1.5029 | 1.5029 | 1.5029 |
| | Max(V(out)) | 0.000 | 0.000 | 0.000 | | -1.5029 | -1.5029 | 0.000 | 0.000 | 0.000 |

Figure 18. Correct limiting circuit operation

## 6.4. Zener diode characteristics

To see if we have modeled the diode correctly, we need to simulate the forward bias current-voltage characteristic and compare it with the one in figure 6. In the configuration where the diode is in the circuit we cannot display this characteristic, therefore we create a circuit that allows us to do this, as seen in figure 19.

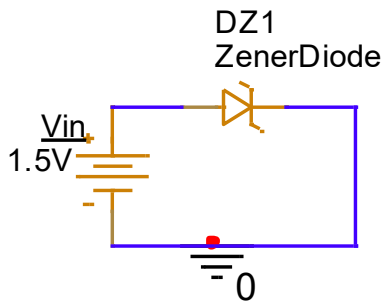


Figure 19. Test circuit for Zener diode forward bias

To simulate the forward bias characteristic, we will perform a DC Sweep analysis by linearly sweeping the Vin source in the test circuit between 0V and 1.4V in steps of 0.1 (figure 20), and display the current through the diode (figure 21).

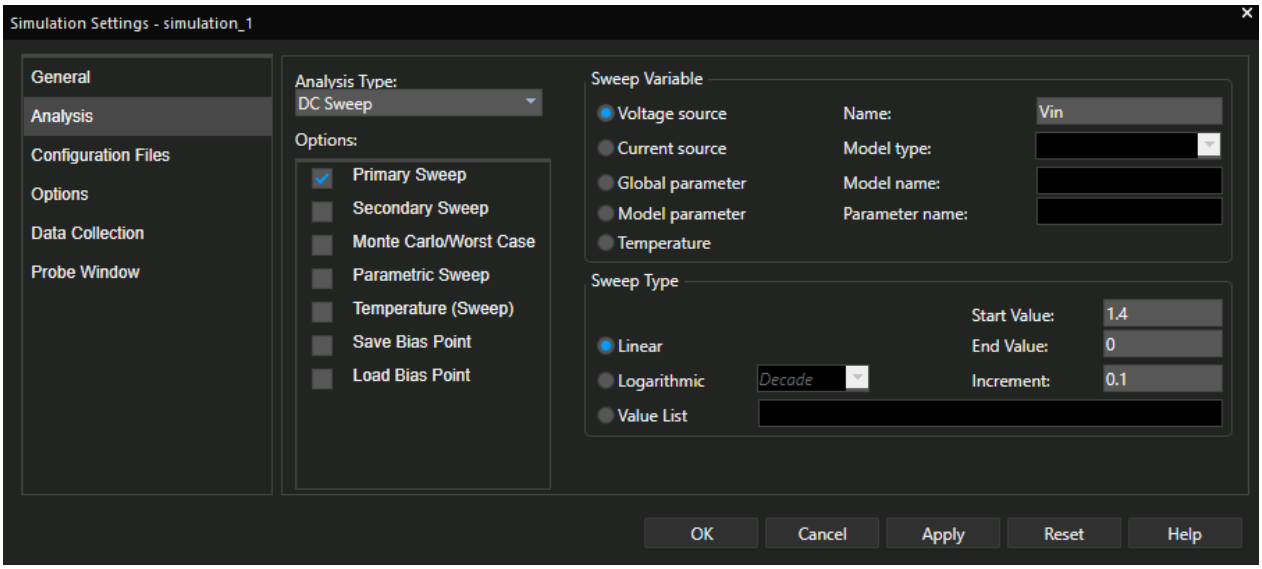


Figure 20. Simulation setup for DC Sweep

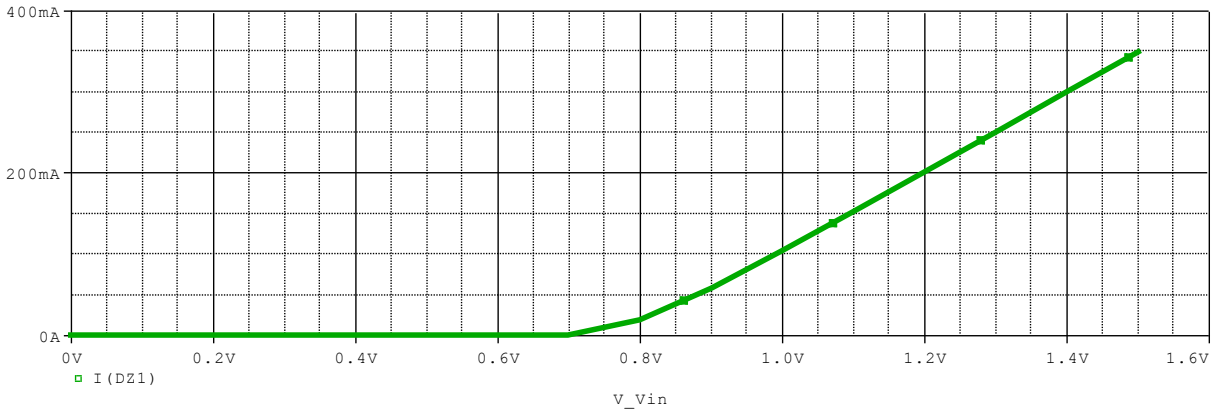


Figure 21. Forward bias characteristic of the Zener diode

To check the diode in reverse bias, we need to observe if the diode complies with the datasheet current at the stabilization voltage. For this, we will need to display both the voltage and the current through the diode, along with a DC Sweep analysis on the 25V input voltage source. We will linearly sweep the voltage source in the 0-25V range, with a step of 0.1, as can be seen in figure 22.

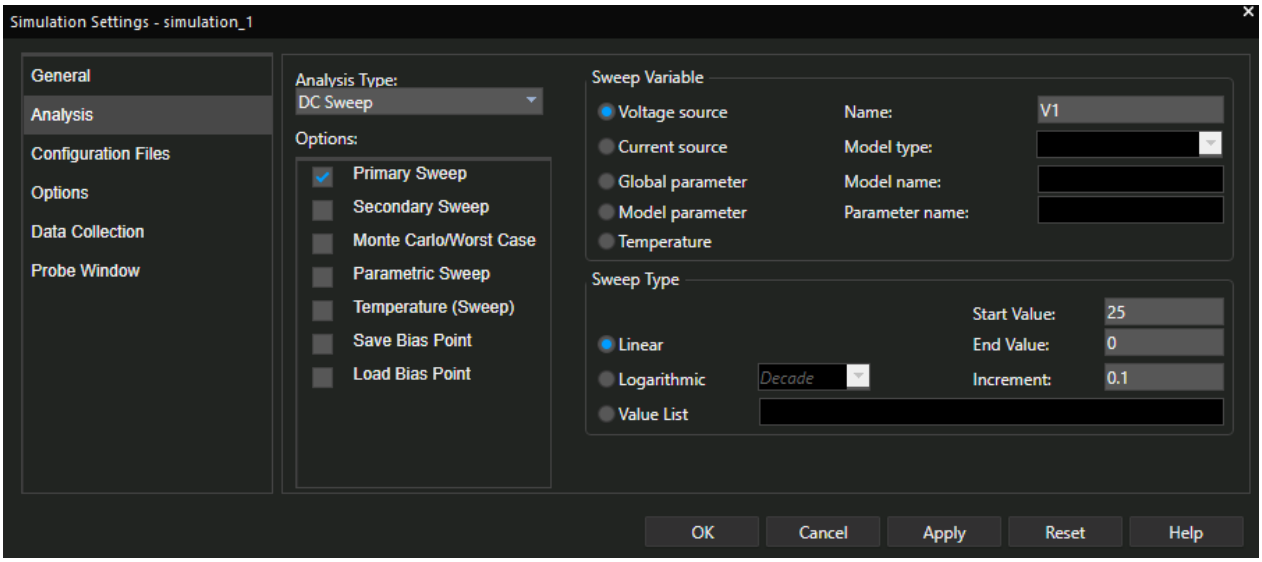


Figure 22. Simulation setup for DC Sweep analysis

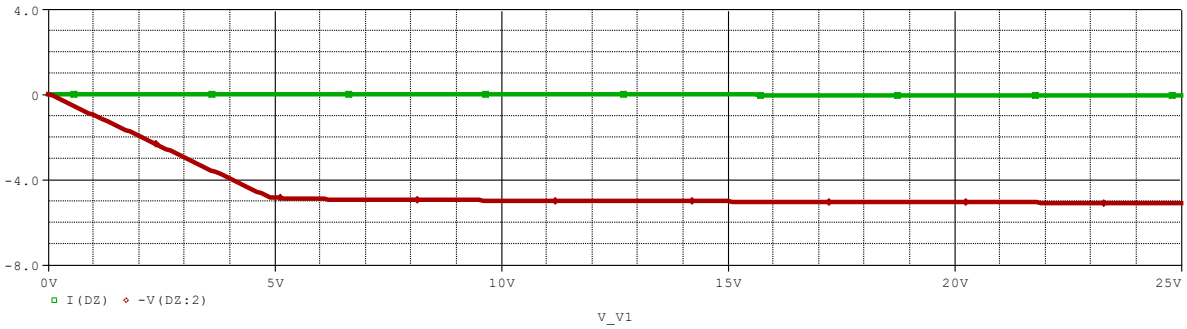


Figure 23. Reverse bias characteristic of the Zener diode

Because we have a small current, measured in milliamps, we cannot really tell its value in relation to the voltage measured in volts on the graph. Therefore, to confirm the correctness of the simulation, we positioned two cursors and their results are displayed in the table in figure 24.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Trace Color | Trace Name | Y1 | Y2 | Y1 - Y2 | | Y1(Cursor1) - Y2(Cursor2) | | 504.510u | | |
| | | X Values | 24.802 | 25.000 | -198.020m | | Y1 - Y1(Cursor1) | Y2 - Y2(Cursor2) | Max Y | Min Y | Avg Y |
| | CURSOR 1,2 | I(DZ) | -49.003m | -49.508m | 504.510u | | 0.000 | 0.000 | -49.003m | -49.508m | -49.256m |
| | | -V(DZ:2) | -5.1001 | -5.1015 | 1.4038m | | -5.0511 | -5.0520 | -5.1001 | -5.1015 | -5.1008 |

Figure 24. Cursor table for characteristic corectness

As we can see in the table above, the characteristic works correctly because at the stabilization voltage we have a current of 49mA through the diode, exactly as in the datasheet.

## 6.5. Worst case analysis

To see how the circuit behaves in the worst case, when we consider resistor`s tolerances, both LOT and DEV, we performed the Worst Case analysis for the output voltage, as shown in Figure 25, to see between which values the minimum and maximum values change. To achieve this, we performed a main Transient analysis for 100ms and checked the Worst Case option with a number of 5 runs (number of components with tolerance + 1).

To begin with, we will leave the output voltage at 7V and perform the Worst Case analysis both High, when the resistors have a value above the nominal value (+1%) (figure 27), and Low, when the resistors have a value below the nominal value (-1%) (figure 28), using the More Settings option in the simulation profile (figure 26).

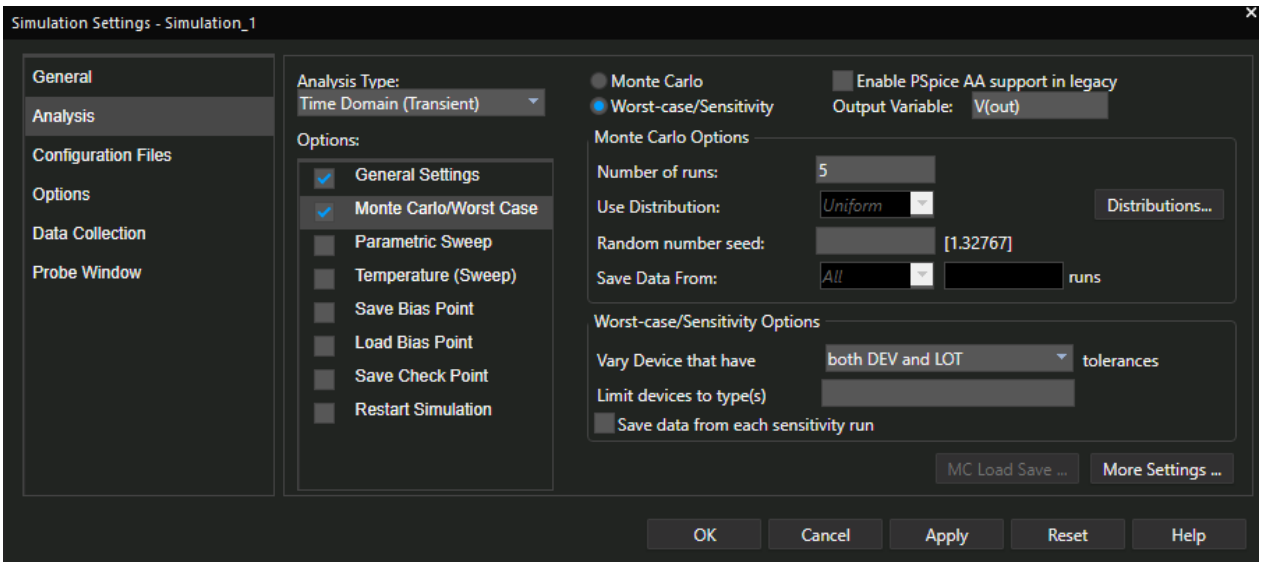


Figura 25. Simulation setup for Worst Case analysis

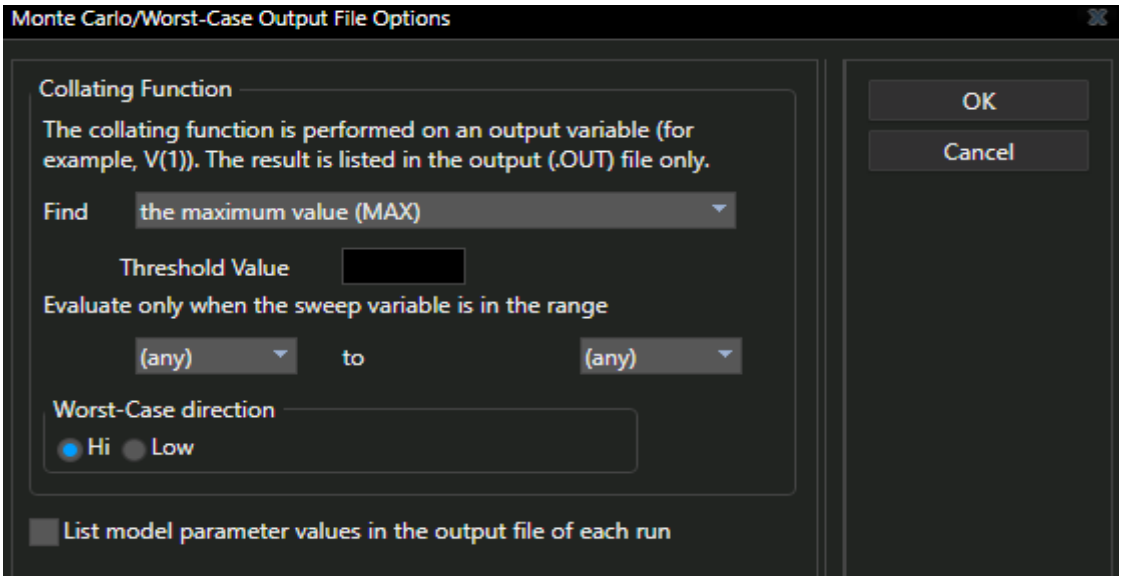


Figure 26. Setting worst case direction

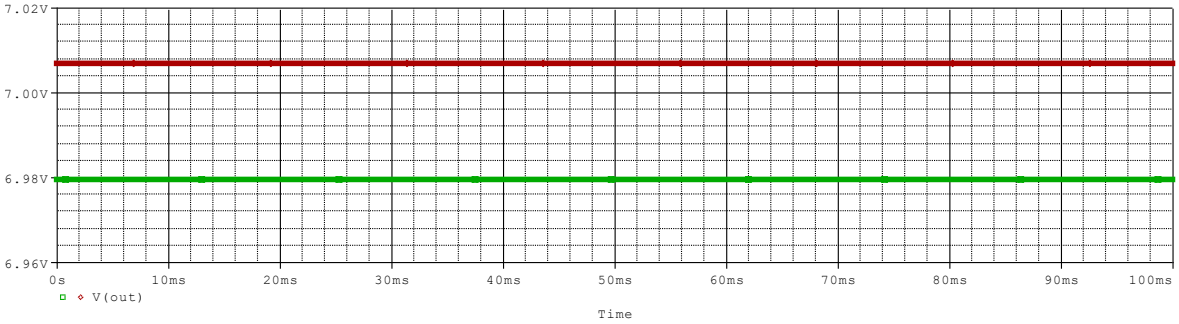


Figure 27. Worst Case $Vout_min$ High

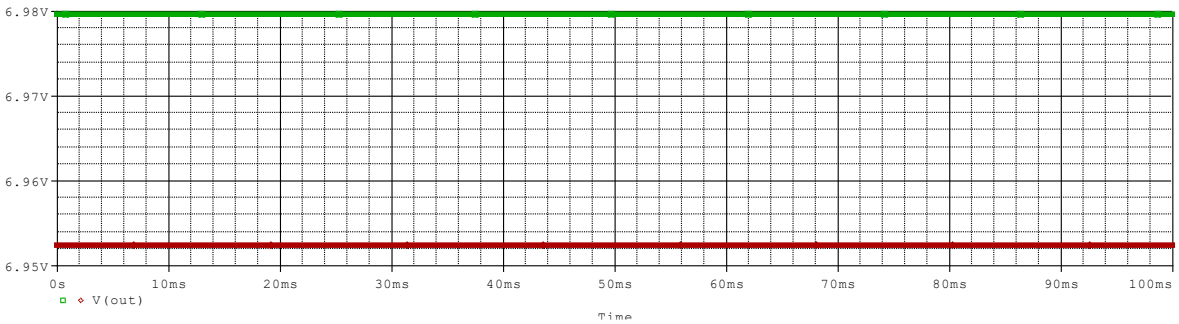


Figure 28. Worst Case $Vout_min$ Low

Now, we move the output voltage to 20V and redo the Worst Case analysis both High, when the resistors have a value above the nominal value (+1%) (figure 29), and Low, when the resistors have a value below the nominal value (-1%) (figure 30).

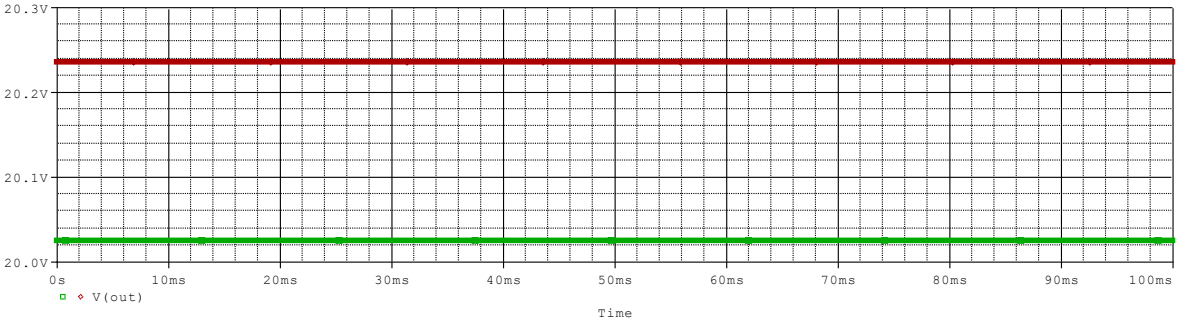


Figura 29. Worst Case $Vout_max$ High

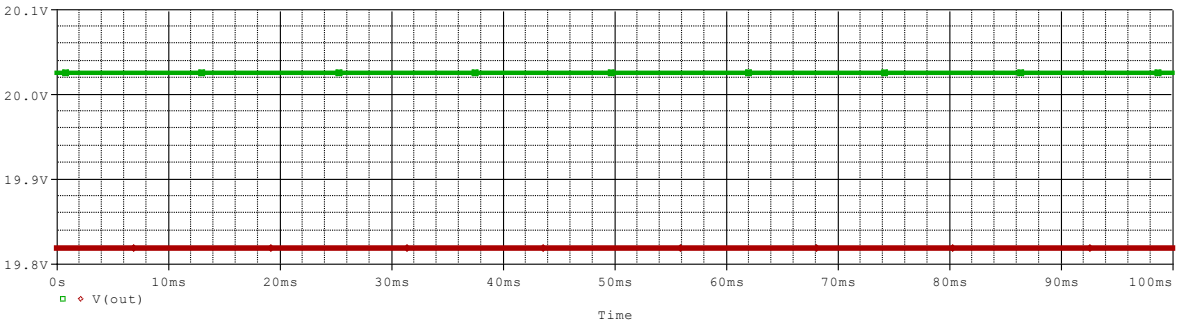


Figura 30. Worst Case $Vout_max$ Low

As we can see from the four simulations, the stabilization of the output signal is not significantly affected by the 1% resistor tolerances.

## 6.6. Monte Carlo analysis

To find out how the circuit behaves taking into account the imperfections of the components used, we performed a Monte Carlo analysis for the minimum and maximum output voltage. To achieve this, we perform a main Transient analysis and, in the simulation profile, we check the Monte Carlo option and set a number of 50 runs, as in figure 31.

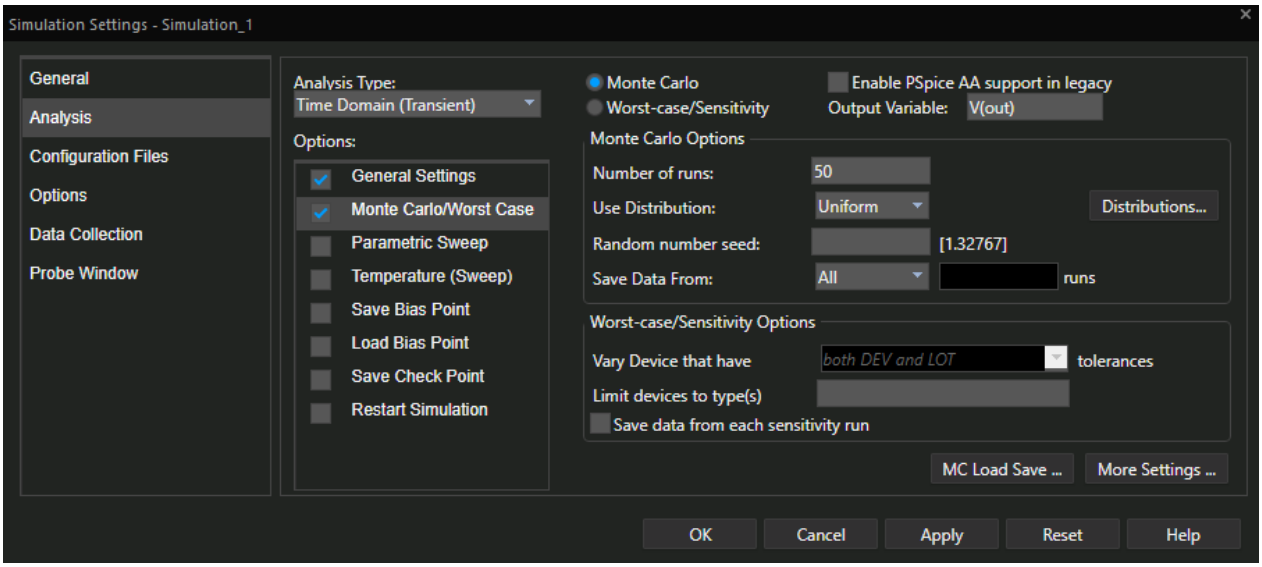


Figure 31. Simulation setup for Monte Carlo analysis

Next, to see the simulation results, we select the performance analysis after running the simulation and display the output voltage on the histogram, as we can see in the figures below.

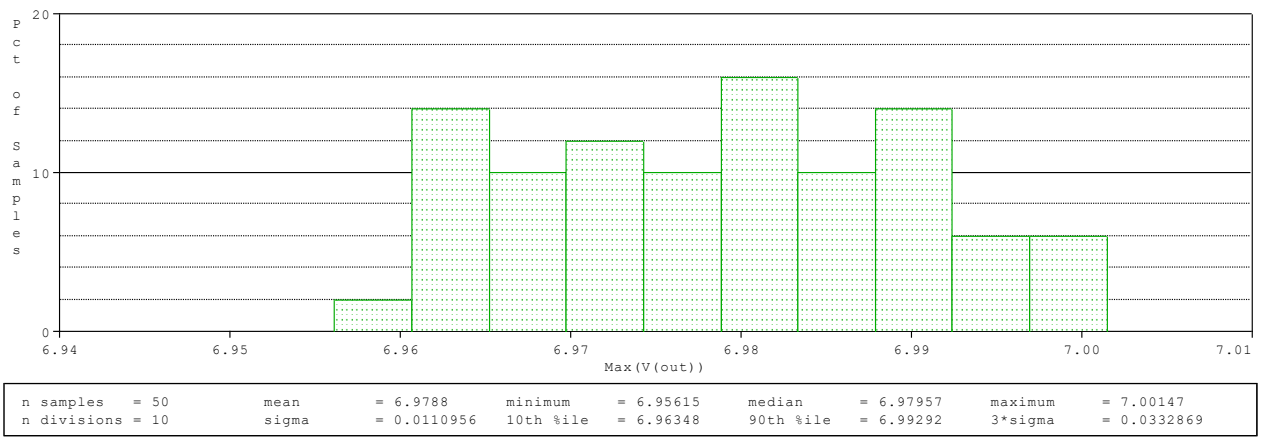


Figure 32. Monte Carlo analysis histogram for $Vout$ min

From Figure 32 we can see that the minimum output voltage, in most cases, is in the range of 6.96V-6.99V, which would not affect the operation of the circuit. However, there is a considerable chance that the minimum output voltage will be located around 6.98V.

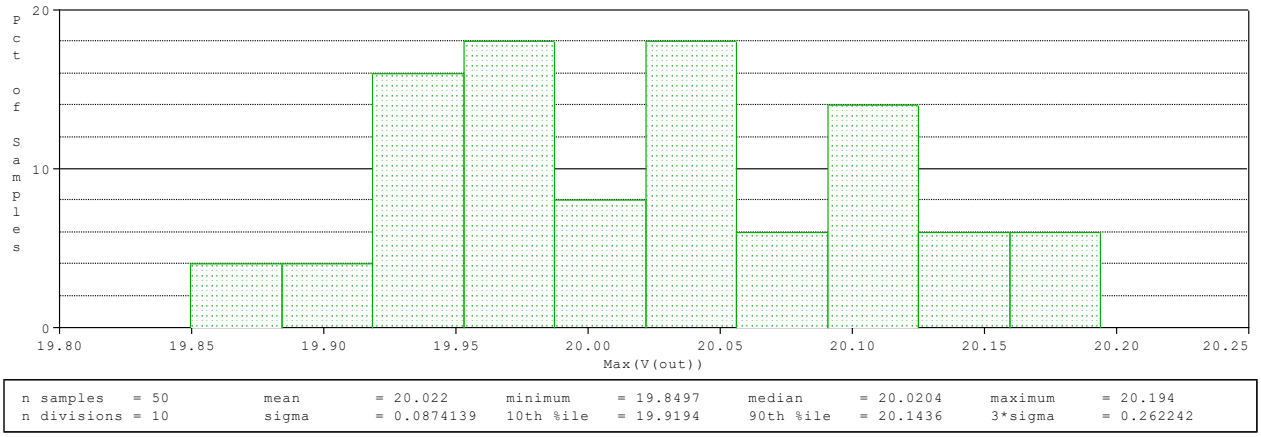


Figure 33. Monte Carlo analysis histogram for $Vout$ max

From figure 33 we can observe that, in most cases, the maximum output voltage is found between 19.92-20.12V. There is also a high chance that the maximum output voltage will be located between 19.95-19.98V and 20.02-20.06V, which is still an acceptable value for stabilization, taking into account the fact that we have an average voltage value of 20.02V.

## 6.7. Temperature analysis

Given that it was not mentioned under what conditions the stabilizer will be used, I decided to also perform a temperature analysis to see if the circuit stabilizes correctly, regardless of the temperature at which it is used.

For this, we performed a main Transient analysis that simulates 100ms, along with the Temperature Sweep option, where we entered 5 different temperature values for which to test stabilization: 0degC, 10degC, 20degC, 30degC and 40degC, as can be seen in figure 34. After running the simulation, we displayed the output voltage (figure 35 and figure 36).

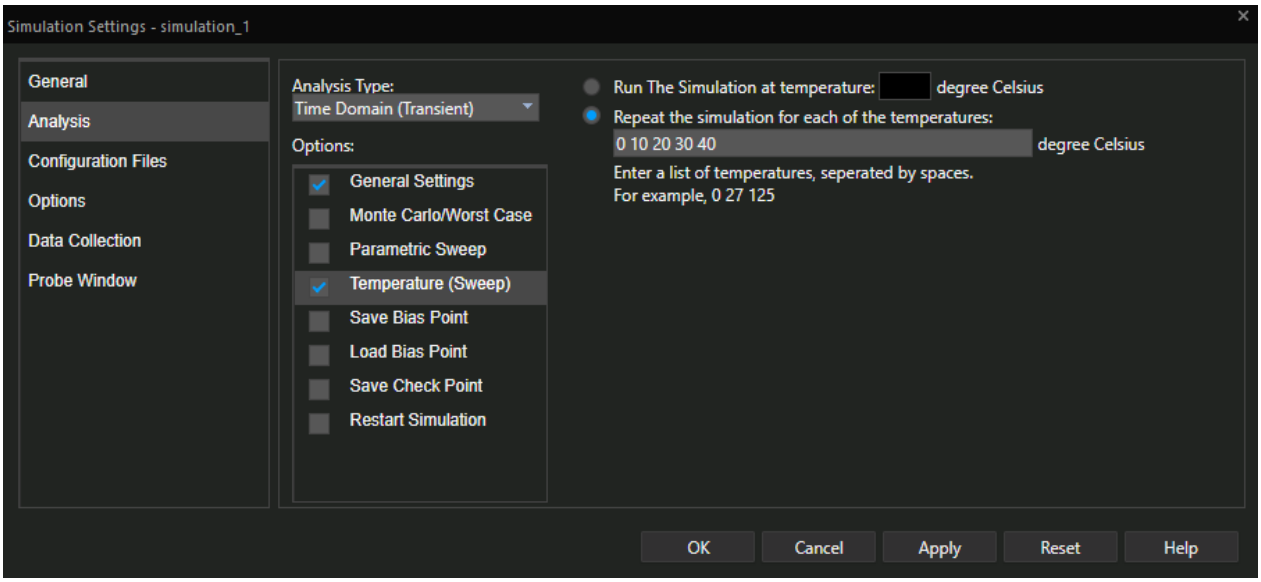


Figure 34. Simulation setup for temperature sweep

Figure 35. Output voltage at 0ºC, 10ºC, 20ºC, 30ºC and 40ºC

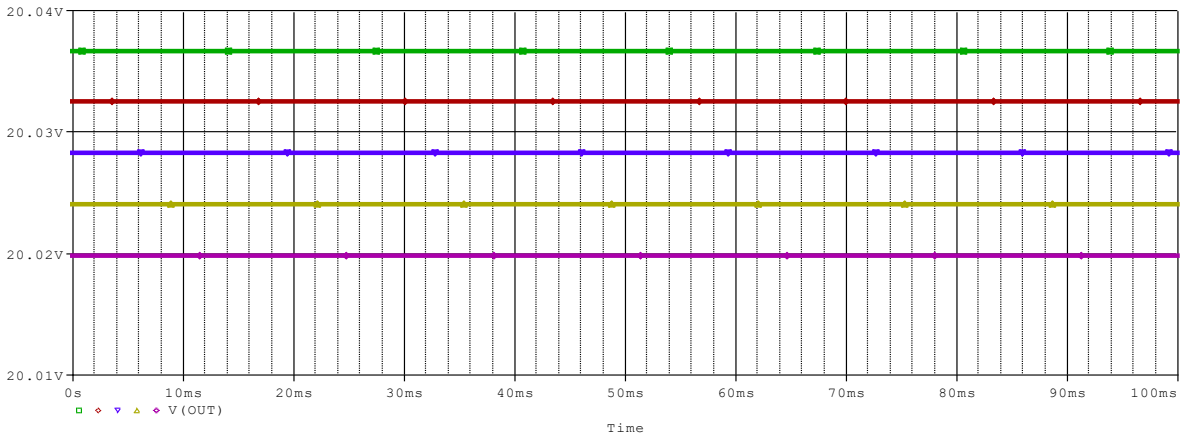


Figure 36. Output voltage at 0ºC, 10ºC, 20ºC, 30ºC and 40ºC

We observe from the two simulations that the stabilization increases with increasing temperature, but the increase is very small between temperatures, therefore, we can say that the stabilizer would work correctly between 0degC and 40degC.

## 6.8. Voltage transfer characteristic

To see how the output voltage varies across the input voltage, I decided to perform a linear DC Sweep analysis on the input source V1, over a range of 0-25V with a step of 0.1, as we can see in figure 39. After running the simulation, we displayed the output voltage in order to visualize the characteristic.

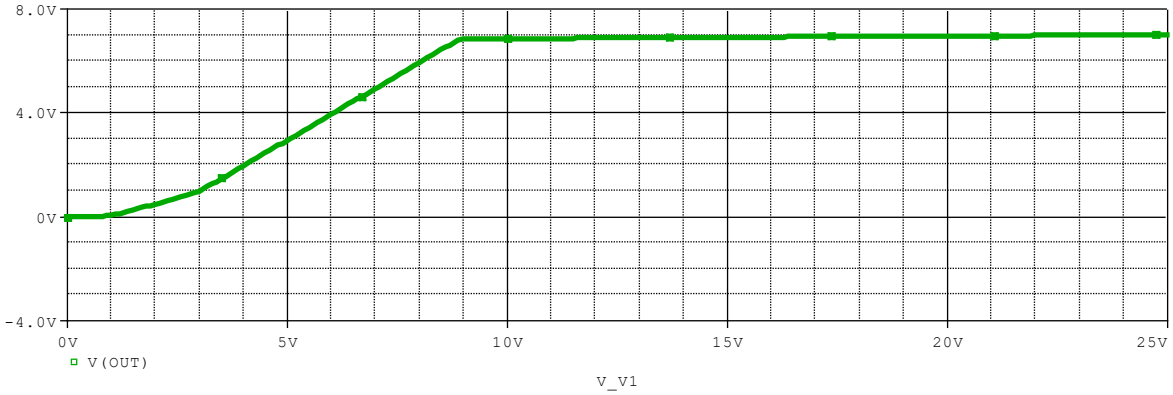


Figure 37. Voltage transfer characteristic for $Vout_min$

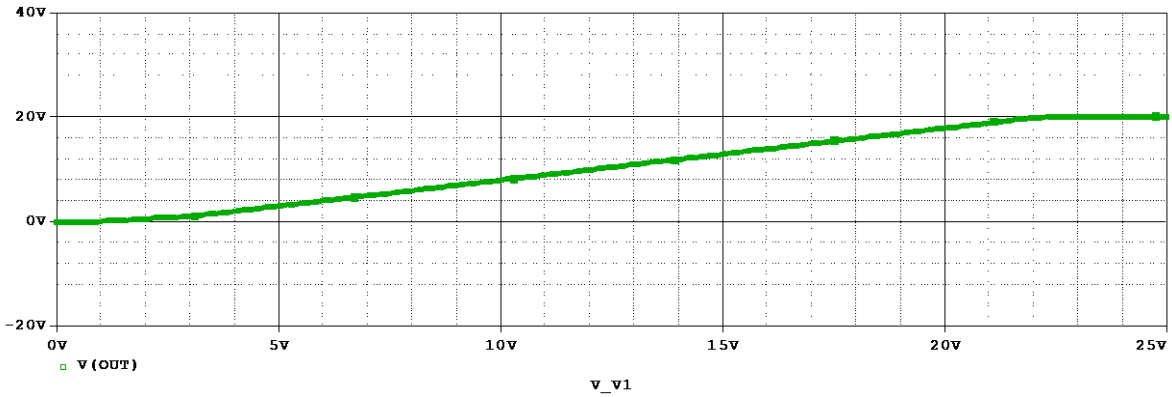


Figure 38. Voltage transfer characteristic for $Vout_max$

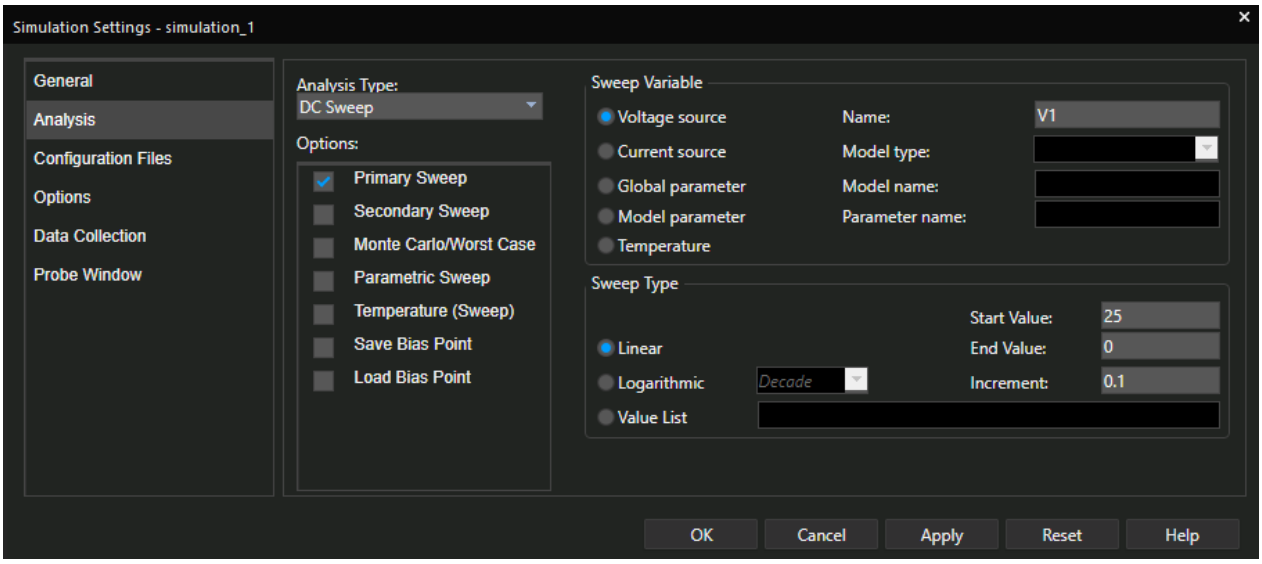


Figure 39. Simulation profile

As we can see from figure 37 and 38, the output voltage increases with input voltage, but the output voltage is limited when it reaches the stabilization value, both minimum and maximum.

# 7. Bill of materials

As a final step, I decided to find out how much it would cost to physically implement the circuit. For this, I used the Mouser Electronics website to find each needed component.

## 7.1. Resistors

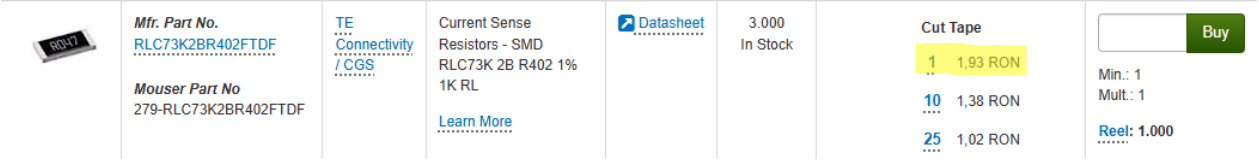


Figure 40. 402Ω resistor

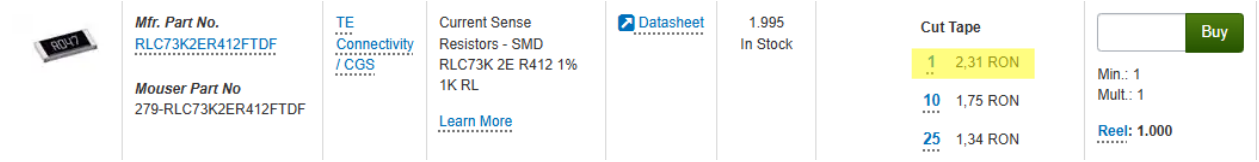


Figure 41. 0.412Ω resistor

Figure 42. 11.3kΩ resistor

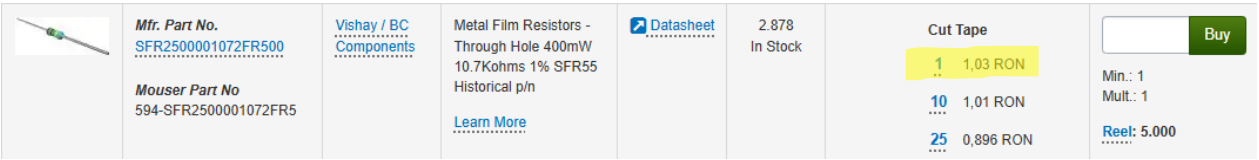


Figure 43. 10.7kΩ resistor

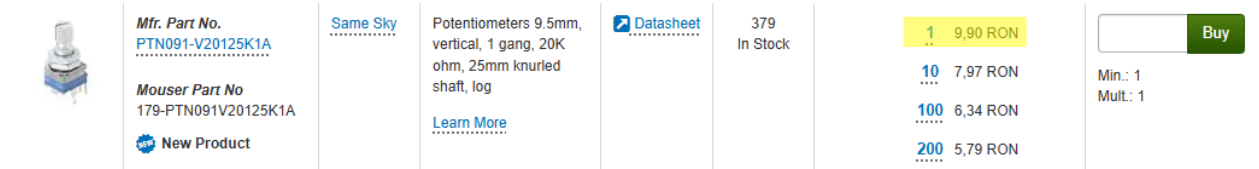


Figure 44. 20kΩ potentiometer

Total: 16.58 RON.

## 7.2. Zener diode

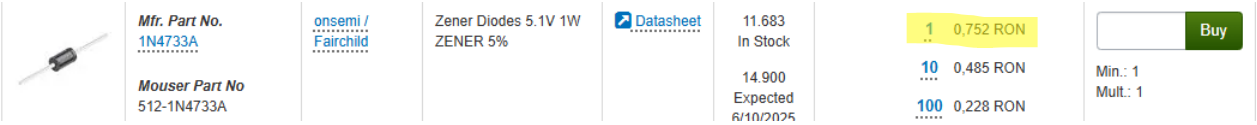


Figure 45. 1N4733A Zener diode

## 7.3. Operational amplifier

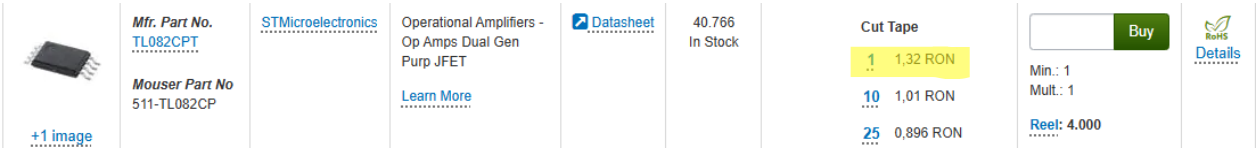


Figure 46. TL082 operational amplifier

## 7.4. Regulation transistor

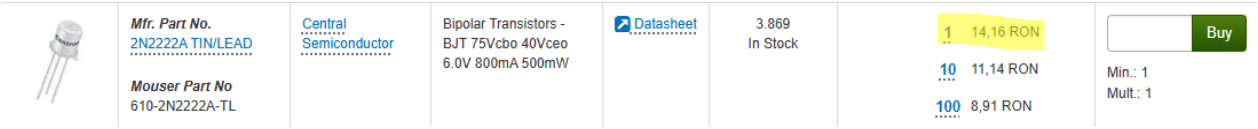


Figure 47. Q2N2222 transistor

## 7.5. Protection transistor

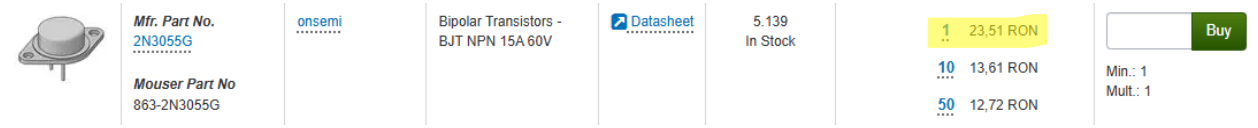


Figura 48. Q2N3055 transistor

Total component price: 55.32 RON.

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[6] Regulation transistor datasheet:
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[6] Protection transistor datasheet:
https://www.farnell.com/datasheets/4396245.pdf